

IKEv1 aggressive mode (edge case)

Analysed 29/09/2026, 02:44:59 · 1 tunnel · model bundle v1 (a88b2a4) · contract antardrishti.result/1

Summary

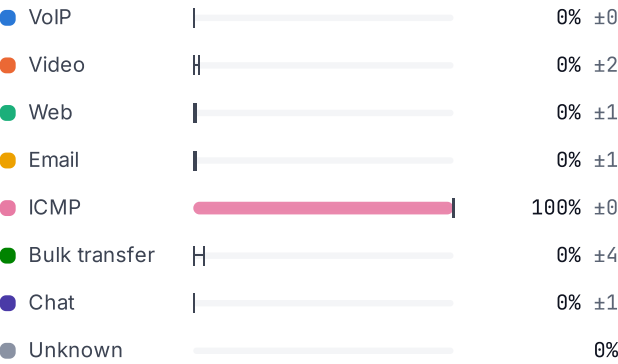
Overall risk	Tunnels	Analysis time	Packets
39	1	0.3 s	90
Inputs: ikev1-aggressive.pcap.zst			

Tunnel #0: 172.31.1.10 → 172.31.2.10

Handshake full · 84 ESP packets · direction from ikev1 first message · risk 39 (medium)

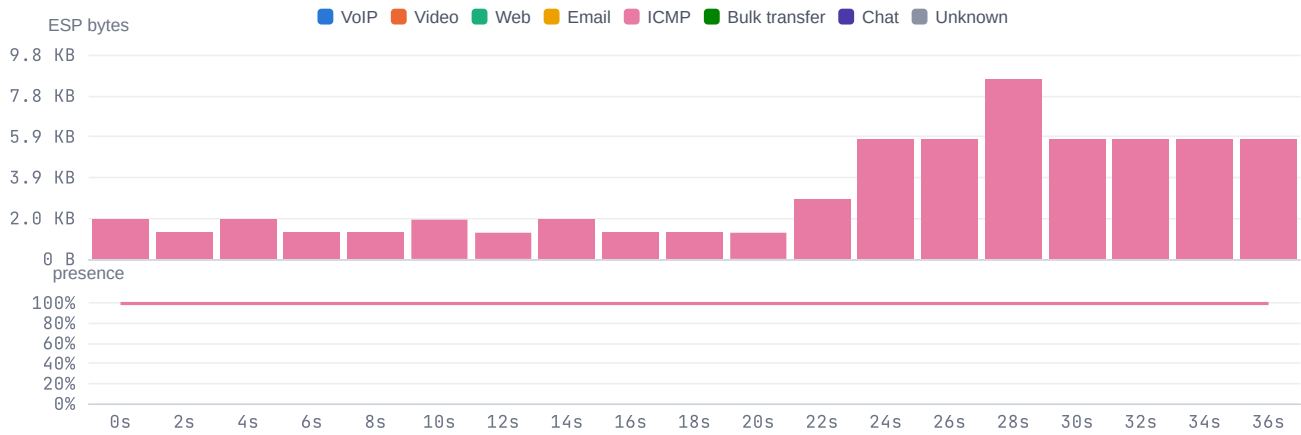
Fact	Value	Confidence	Source
handshake_status	full	100%	 observed
outer_family	ipv4	100%	 observed
nat_t	no	100%	 observed
ike_version	1	100%	 read from IKE
ike_exchanges	QUICK: 3 · AGGRESSIVE: 3	100%	 read from IKE
ike_retransmissions	0	100%	 observed
ike_encryption	aes-cbc-256	100%	 read from IKE
ike_integ	hmac-sha256-128	100%	 read from IKE
ike_dh	ecp256	100%	 read from IKE
ikev1_mode	aggressive	100%	 read from IKE
ikev1_auth	psk	100%	 read from IKE
ike_lifetime_s	15,850	100%	 read from IKE
rekeys	child: 0 · ike_sa: 0	100%	 observed
rekey_times_s	child: none · ike_sa: none	100%	 observed
spi_first_seen_s	ca151383: 0.092417 · cf706eb1: 0.092333	100%	 observed
capture_span_s	37.156	100%	 observed
ah_packets	0	100%	 observed
esp_plaintext_share	0	100%	 observed
esp_duplicate_seq	0	100%	 observed
esp_suite	AES-GCM-16 (ICV 16)	100%	 inferred
mode	tunnel	99%	 inferred
pfs	not determinable	 n/d	 inferred

Traffic shares



App	Bytes	Error bar	Active time
VoIP	0%	0-0%	0%
Video	0%	0-2%	0%
Web	0%	0-1%	0%
Email	0%	0-1%	0%
ICMP	100%	100-100%	100%
Bulk transfer	0%	0-4%	0%
Chat	0%	0-1%	0%
Unknown	0%	-	-

Window timeline



Findings

Check	Verdict	Finding, evidence and standard	Confidence
IKEV1-AGGR tunnel 0	 Fail  High	IKEv1 aggressive mode sends the identity in the clear and, with a pre-shared key, a hash an attacker can crack offline ikev1_mode = aggressive (read from IKE) NIST SP 800-77r1 (Guide to IPsec VPNs) Fix: Disable aggressive mode (or move to IKEv2).	100%
IKE-VERSION tunnel 0	 Fail  Medium	The tunnel uses IKEv1, which NIST SP 800-77r1 recommends against; use IKEv2 ike_version = 1 (read from IKE) NIST SP 800-77r1 (Guide to IPsec VPNs); RFC 8247 (IKEv2 algorithm requirements) Fix: Migrate the tunnel to IKEv2.	100%
METADATA tunnel 0	 Warn  Medium	A passive observer can tell the traffic inside apart: icmp (100% of bytes, presence 99%) identifiable = icmp (100% of bytes, presence 99%) (inferred) RFC 4303 section 2.7 (traffic flow confidentiality padding); NIST SP 800-77r1 (Guide to IPsec VPNs) Fix: Enable TFC padding (RFC 4303 section 2.7) or dummy traffic where hiding activity matters.	99%
AUTH tunnel 0	 Warn  Low	IKEv1 authenticates with a pre-shared key; its strength cannot be seen ikev1_auth = psk (read from IKE) NIST SP 800-77r1 (Guide to IPsec VPNs); RFC 8247 (IKEv2 algorithm requirements) Fix: Prefer certificate (RSA-PSS or ECDSA) authentication; if a PSK is used, make it long and random.	100%
ESP-KEYLEN tunnel 0	 Not determinable	The AES key size never shows in ESP (same IV, ICV and block padding for 128 and 256-bit keys) and the child SA proposal is encrypted inside IKE_AUTH RFC 8221 (ESP and AH algorithm requirements)	 n/d
IKE-PRF tunnel 0	 Not determinable	The IKE_SA_INIT exchange is not in the capture (or IKEv1) RFC 8247 (IKEv2 algorithm requirements)	 n/d
LIFETIME-CHILD tunnel 0	 Not determinable	Fewer than two child SA rekeys in the capture, so the lifetime is not visible NIST SP 800-77r1 (Guide to IPsec VPNs)	 n/d
PFS tunnel 0	 Not determinable	PFS only shows in a child SA rekey, and none is in the capture pfs = not determinable (inferred) NIST SP 800-77r1 (Guide to IPsec VPNs); RFC 7296 (IKEv2)	 n/d
REPLAY-WINDOW tunnel 0	 Not determinable	The replay window and ESN are negotiated and enforced inside the gateways; the wire only shows the low 32 sequence bits RFC 4303 section 3.4.3	 n/d
REPLAY tunnel 0	 Info	No repeated ESP sequence numbers seen on the wire (an injected replay may not cross this vantage point) esp_duplicate_seq = 0 (observed) RFC 4303 section 3.4.3 (sequence number verification)	100%

Check	Verdict	Finding, evidence and standard	Confidence
AH-ONLY tunnel 0	 Pass	No AH packets ah_packets = 0 (observed) RFC 8221 (ESP and AH algorithm requirements); NIST SP 800-77r1 (Guide to IPsec VPNs)	100%
COOKIE tunnel 0	 Pass	No cookie challenge cookie = no (read from IKE) RFC 7296 (IKEv2)	100%
ESP-NULL tunnel 0	 Pass	ESP payloads look encrypted (0.0% resemble cleartext by chance) esp_plaintext_share = 0 (observed) RFC 8221 (ESP and AH algorithm requirements); NIST SP 800-77r1 (Guide to IPsec VPNs)	100%
ESP-SUITE tunnel 0	 Pass	ESP uses AES-GCM-16 (ICV 16) esp_suite = AES-GCM-16 (ICV 16) (inferred) RFC 8221 (ESP and AH algorithm requirements); NIST SP 800-77r1 (Guide to IPsec VPNs)	100%
IKE-DH tunnel 0	 Pass	The key exchange uses ecp256 ike_dh = ecp256 (read from IKE) RFC 8247 (IKEv2 algorithm requirements); NIST SP 800-77r1 (Guide to IPsec VPNs)	100%
IKE-ENC tunnel 0	 Pass	IKE SA uses aes-cbc-256 ike_encryption = aes-cbc-256 (read from IKE) RFC 8247 (IKEv2 algorithm requirements); NIST SP 800-77r1 (Guide to IPsec VPNs)	100%
IKE-INTEG tunnel 0	 Pass	IKE SA integrity is hmac-sha256-128 ike_integ = hmac-sha256-128 (read from IKE) RFC 8247 (IKEv2 algorithm requirements)	100%
IKE-KEYLEN tunnel 0	 Pass	The IKE SA uses a 256-bit AES key ike_key_bits = 256 (read from IKE) RFC 8247 (IKEv2 algorithm requirements); NIST SP 800-77r1 (Guide to IPsec VPNs)	100%
LIFETIME-IKE tunnel 0	 Pass	The IKE SA is replaced within 15850 s ike_lifetime_s = 15,850 (read from IKE) NIST SP 800-77r1 (Guide to IPsec VPNs)	100%
MODE tunnel 0	 Pass	The tunnel is in tunnel mode (inner addresses hidden) mode = tunnel (inferred) NIST SP 800-77r1 (Guide to IPsec VPNs)	99%
NEGOTIATION tunnel 0	 Pass	Handshake status: full handshake_status = full (observed) RFC 7296 (IKEv2)	100%
NOTIFY tunnel 0	 Pass	No error notifies in the cleartext IKE messages error_notifies = none (read from IKE) RFC 7296 (IKEv2)	100%
RETRANSMIT tunnel 0	 Pass	0 IKE retransmissions ike_retransmissions = 0 (observed) RFC 7296 (IKEv2)	100%

Threat matrix

Threat	Likelihood	Impact	Tunnels	Linked findings
Offline recovery of keys or the pre-shared key T-KEY	4	4	0	IKEV1-AGGR#0, AUTH#0
IKE downgrade or man-in-the-middle T-MITM	3	4	0	IKE-VERSION#0
Traffic analysis and metadata exposure T-META	4	2	0	IKEV1-AGGR#0, METADATA#0

Model and methodology

- Model bundle v1, commit a88b2a4f3f6126af42900121aab056a8166b15b6, 17 LightGBM models (3 config, 7 presence, 7 share), seed 26006; data releases p0-data, p0-labels, p0s-data, p0s-labels, p1-data, p1-labels, p1-whatsapp, p1-whatsapp-labels.
- Parser: outer IP fragments reassembled; packets grouped into tunnels by address pair, NAT-T ports, IKE SPIs and ESP SPIs; IKE read in the clear where captured.
- Config models read ESP length residues and minimum sizes (suite, mode) and CREATE_CHILD_SA sizes (PFS). Traffic models classify 2-second windows from sizes, timing, bursts and FFT; shares are byte-weighted, absent apps zeroed, rounded to 100 by largest remainder.
- Error bars: out-of-fold session share error per app and share range, clipped to 0-100.
- Rules: 23 checks against RFC 8221, RFC 8247, NIST SP 800-77r1, RFC 7296 and RFC 4303. Risk: weights critical 40, high 20, medium 8, low 3, info 0 times confidence, capped at 100, a confident critical sets a floor of 90; overall is the worst tunnel.

Passive analysis of the outer capture only: no traffic was decrypted and no keys were used. Facts marked "inferred" come from models trained on the Antardrishti dataset; their confidence is calibrated, and findings below 80% confidence are worded "likely".